WAP to show that JNI (Java Native Interface) classes is generated by JVM for 1D array

The JNI uses the Java VM's representation of type signatures. The following table shows these type signatures.

Java VM Type Signatures

Type Signature	Java Type
Z	boolean
B	byte
C	char
S	short
I	int
J	long
F	float
D	double
L fully-qualified-class ;	fully-qualified-class
[type	type[]
(arg-types) ret-type	method type

Those are classes generated by JVM at runtime for the respective data types.

```
package com.ravi.basic;

public class ArrayJni
{
    public static void main(String[] args)
    {
        byte bytearr[] = new byte[5];
        IO.println(bytearr.getClass().getName());

        short shortarr[] = new short[5];
        IO.println(shortarr.getClass().getName());

        int intarr[] = new int[5];
        IO.println(intarr.getClass().getName());
        IO.println(intarr.getClass().getSuperclass());//extends object implements Cloneable
        IO.println(intarr.getClass().getInterfaces());
        IO.println(intarr.getClass().getModifiers());
        IO.println(intarr.getClass().getModifiersExt());


        long longarr[] = new long[5];
        IO.println(longarr.getClass().getName());


        float floatarr[] = new float[5];
        IO.println(floatarr.getClass().getName());

        double doublearr[] = new double[5];
        IO.println(doublearr.getClass().getName());


        char chararr[] = new char[5];
        IO.println(chararr.getClass().getName());


        boolean booleanarr[] = new boolean[5];
        IO.println(booleanarr.getClass().getName());


        String [] title = {"Prof", "Sir", "Mam"};                    [  -> 1D Array
        IO.println(title.getClass().getName());                      L  -> The classes passed by JVM

    }
}
```

Q1) Different ways to create 1D array and fetching the elements using ordinary for loop /forEach loop and tabbing[].

public static String toString(int[]arr) ;

Arrays class has provided a predefined static method called **toString()** to print the **1D array** data into String format.

It is available from JDK 1.9V

```
package com.ravi.array;

import java.util.Arrays;

public class ArrayDemo1 {

    public static void main(String[] args)
    {
        IO.println("Approach 1");

        int size = Integer.parseInt(IO.readln("Enter the size of the array :"));
        int []arr1 = new int[size];

        //initializing the array elements using for loop

        for(int i=0; i<size; i++)  //i=5
        {
            IO.print("Enter the element of the array at "+i+" index :");
            arr1[i] = Integer.parseInt(IO.readln());
        }

        IO.println("Printing the Array elements :");

        for(int i=0; i<size; i++)
        {
            IO.println(arr1[i]);
        }

        IO.println("Approach 2");

        int []arr2 = new int[] {100, 200, 300, 400};

        for(int element : arr2)
        {
            IO.println(element + "\t");
        }

        IO.println();
        IO.println("Approach 3");

        int []arr3 = {1000, 2000, 3000, 4000};
        IO.println(Arrays.toString(arr3));

    }

}
```

***Q2) WAP to print the array elements in reverse order. (User input)

```
package com.ravi.array;

public class ArrayDemo2 {

    public static void main(String[] args)
    {
        int size = Integer.parseInt(IO.readln("Enter the size of the array :"));

        int []arr = new int[size];  //10  20   30   40   50

        for(int i=0; i<arr.length; i++)
        {
            IO.print("Enter the element of the array at "+i+" index :");
            arr[i] = Integer.parseInt(IO.readln());
        }

        IO.println("Original Array :");
        for(int element : arr)
        {
            IO.print(element+" \t");
        }

        IO.println();

        IO.println("Array elements in reverse order :");

        for(int i=arr.length-1; i>=0; i--)
        {
            IO.println(arr[i]);
        }

    }

}
```

Q3) WAP to find the the sum of all elements of the array.

```
package com.ravi.array;

import java.util.Arrays;

public class ArrayDemo3 {

    public static void main(String[] args)
    {
        int []arr = {30,20,30,40,50}; //Brute force approach

        int sum = 0;

        for(int element : arr)
        {
            sum = sum + element;
        }

        IO.println("Sum of array elements are :"+sum);

        IO.println("Java 8 Approach");
        sum = Arrays.
            stream(arr).
            sum();
        IO.println("Sum of array elements are :"+sum+" using Java 8");
    }
}
```

Q4) WAP to calculate the student marks and find out the average. Assignment

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Q5) WAP to find the minimum element in the Array. (maximum also)

```
package com.ravi.array;

import java.util.Arrays;

class FindMinimum
{
    public static int getMinimumElement(int []arr)
    {
        int min = arr[0];

        for(int element : arr)
        {
            if(element < min)
            {
                min = element;
            }
        }

        return min;
    }
}

public class ArrayDemo4 {

    public static void main(String[] args)
    {
        int values[] = {30, 2, 6, 1, 5, 4};
        IO.println("Original Array is :"+Arrays.toString(values));

        int minElement = FindMinimum.getMinimumElement(values);
        IO.println("Min element is :"+minElement);

    }

}
```